Strategy-Relative Regime Switching as a Verifiable RLVR Environment for Financial Planning

Anonymous Authors

Submitted to NeurIPS 2026

Abstract

We introduce the first verifiable reinforcement learning (RLVR) environment for financial planning, designed for large language model (LLM) post-training. The environment is grounded in the strategy-relative regime switching framework of Bilokon (2026), where an agent makes binary switching decisions under an observable but noisy signal, subject to switching costs. Bellman backward induction on a discretized state space provides exact optimal policies, enabling mathematically rigorous reward computation. A parameter landscape analysis across 1,500 configurations identifies when forward-looking planning diverges from myopic behavior: signal persistence (κ) is the dominant factor, with slow mean-reversion ($\kappa = 0.1$) yielding up to 22% state-level disagreement between optimal and greedy policies. We propose a two-bucket evaluation that separates "easy" decisions (greedy = optimal) from "hard" decisions (greedy $\neq$ optimal). Evaluating Claude Opus 4 on 750 decisions, we find 99.3% accuracy on easy decisions but only 5.9% on hard decisions—the model matches greedy behavior on 94.1% of hard decisions, showing near-zero forward-looking reasoning. The gym is open-source with single-function TRL integration.

1. Introduction

Verifiable reinforcement learning from verifiable rewards (RLVR) has emerged as a powerful paradigm for LLM post-training. DeepSeek-R1 (DeepSeek-AI et al., 2025) demonstrated that GRPO training with verifiable rewards can elicit chain-of-thought reasoning in mathematical problem-solving. Reasoning Gym (Stojanovski et al., 2025) extended this to a diverse suite of puzzles with computable solutions.

However, existing RLVR gyms test **static problem-solving with complete information**: a math problem presents all premises at once, and the model produces a single answer. Many real-world capabilities require **sequential decision-making under uncertainty**, where the agent must act before the full picture is revealed and each action constrains future actions. Consider:

> **Math gym (static):** A trader pays \$0.15 to switch. Expected return is $0.30 \times Z_t$. Signal is $+0.35$. Should the trader enter? A single computation: $0.30 \times 0.35 = 0.105 < 0.15$. Answer: no.
>
> **Our gym (sequential):** Same parameters—but if the signal is persistent (slow mean-reversion), the cumulative returns over many steps will exceed the one-time cost. The correct answer depends on *how long the signal persists.*

This distinction—reasoning about a trajectory versus a single state—is what separates static from sequential planning. We resolve the verification challenge by building on Bilokon (2026), who derives optimal binary regime processes via Bellman dynamic programming. Because the signal process is fully specified and the state space is finite after discretization, backward induction yields exact optimal policies, providing a computable ground truth.

Contributions. (1) The first verifiable RLVR environment for sequential financial planning with exact ground-truth solutions. (2) A parameter landscape analysis across 1,500 configurations characterizing when planning matters. (3) A two-bucket evaluation separating comprehension from planning capability. (4) Evaluation of Claude Opus 4 showing near-perfect comprehension (99.3%) but near-zero planning (5.9% hard accuracy).

---

2. Problem Formulation

2.1 Strategy-Relative Regime Switching

Following Bilokon (2026), we consider a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t=0}^{T}, \mathbb{P})$ with an observable signal process Z_t and a PnL process X_{t+1}. The agent controls a binary regime $s_t \in \{0, 1\}$, maximizing:

$$J(s) = \mathbb{E}\left[\sum_{t=0}^{T-1} s_t \cdot \alpha \cdot Z_t - \lambda \cdot \mathbf{1}\{s_t \neq s_{t-1}\}\right]$$

where $\alpha > 0$ is the signal strength, $\lambda > 0$ is the switching cost, and $s_{-1} = 0$.

2.2 Bellman Recursion and Data Generating Process

The value function $V_t(z, s^-)$ satisfies:

$$V_t(z, s^-) = \max_{a \in \{0,1\}} \left[a \cdot \alpha \cdot z - \lambda \cdot \mathbf{1}\{a \neq s^-\} + \int V_{t+1}(z', a)\, P(dz'|z)\right]$$

with $V_T = 0$. The signal follows an Ornstein-Uhlenbeck process $dZ_t = \kappa(\theta - Z_t)\, dt + \sigma_z\, dW_t$ with $\theta = 0$. The parameter κ controls mean-reversion speed: small κ yields persistent signals (favorable for planning), large κ yields rapidly decaying signals (greedy is near-optimal). The Bellman equation is solved by backward induction on a discretized z-grid with Gauss-Hermite quadrature.

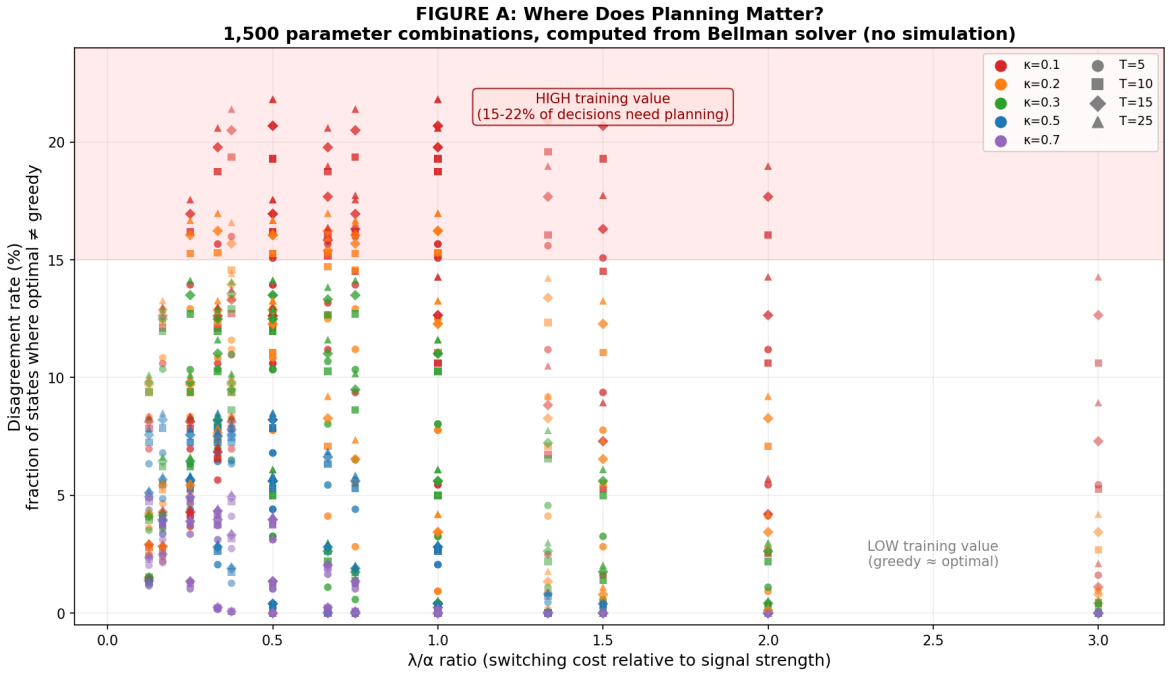


Figure 1: Figure A — Where Planning Matters (1,500 configurations)

3. Gym Architecture

The gym has three components. The **Generator** samples parameters, solves the Bellman recursion, simulates signal trajectories, and computes optimal/greedy/random baselines (50 seeds per configuration). The **Verifier** parses LLM decisions and computes a regret-normalized score:

$$\text{score} = \frac{J_{\text{LLM}} - J_{\text{random}}}{J_{\text{optimal}} - J_{\text{random}}}$$

The **Prompt Manager** enforces the information constraint: at each step t, the LLM receives only Z_t, s_{t-1}, the previous PnL, and the parameters λ, α—never future signals. The gym exposes a TRL-compatible reward function for direct GRPO integration.

---

4. Parameter Landscape Analysis

We conduct a full factorial sweep over $\kappa \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$, $\lambda \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$, $\alpha \in \{0.10, 0.15, 0.20, 0.25, 0.30\}$, $\sigma_z \in \{0.1, 0.2, 0.3\}$, and $T \in \{5, 10, 15, 25\}$, yielding 1,500 configurations solved to exact optimality.

For each configuration, we compute a **disagreement metric**: the fraction of (z, s^-) states where Bellman optimal differs from greedy, weighted by the OU stationary distribution.

Key findings (Figures A, D, J). Signal persistence dominates: at $\kappa = 0.1$, mean disagreement is 13.1%; at $\kappa = 0.7$, it drops to 1.2%. The λ/α ratio has a non-monotone effect, peaking at $\lambda/\alpha \in [0.33, 0.50]$. Longer horizons modestly amplify disagreement. The maximum observed disagreement is approximately 22% of grid states (Figure A); on simulated trajectories, approximately 10% of visited decisions differ.

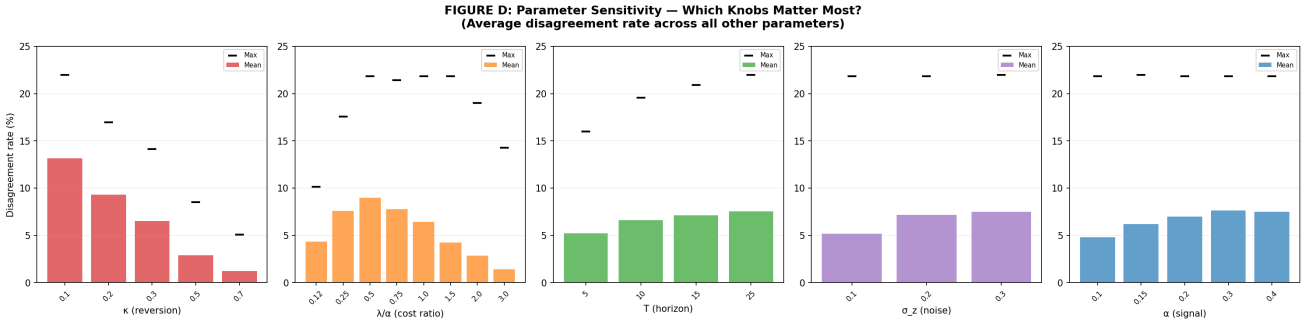


Figure 2: Figure D — Parameter Sensitivity (which knobs matter most)

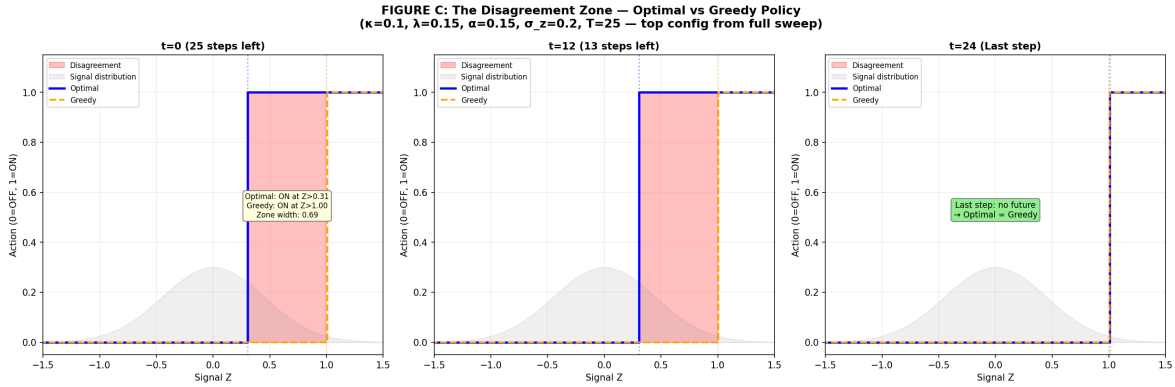


Figure 3: Figure C — The Disagreement Zone (policy comparison at 3 time steps)

5. The Planning Mechanism

5.1 Greedy vs. Optimal Thresholds

The greedy agent switches ON when $\alpha \cdot z > \lambda$, giving threshold $z^{\text{greedy}} = \lambda/\alpha$. At our evaluation configuration ($\lambda = 0.15$, $\alpha = 0.30$), this is $z = 0.50$.

The optimal agent switches ON at a *lower* threshold $z^{\text{opt}} \approx 0.22$, because it amortizes the one-time switching cost over persistent future gains. When κ is small, a signal $Z_t = 0.35$ is likely to remain positive for many steps. The greedy agent sees $0.105 < 0.15$ and stays OFF; the optimal agent recognizes the cumulative gain exceeds the cost.

5.2 The Disagreement Zone

The disagreement zone $\mathcal{D} = \{z : \pi^*(z, s^-) \neq \pi^{\text{greedy}}(z, s^-)\}$ spans approximately $[0.22, 0.50]$ for OFF $\rightarrow$ ON switching (and $[-0.50, -0.22]$ by symmetry). It widens at earlier time steps and collapses to empty at $t = T - 1$, where optimal reduces to greedy (Figure C).

---

6. LLM Evaluation

6.1 Setup and Two-Bucket Methodology

We evaluate Claude Opus 4 on 30 instances at $(\kappa = 0.1, \lambda = 0.15, \alpha = 0.30, T = 25)$, producing 750 decisions (25 per instance). Decisions are partitioned by the solver into **easy** ($n = 682$, greedy = optimal) and **hard** ($n = 68$, greedy $\neq$ optimal). This partition requires no additional LLM queries and separates comprehension from planning.

6.2 Results

Easy bucket: 99.3% accuracy (677/682)—the model correctly evaluates immediate payoffs.

Hard bucket: 5.9% match optimal (4/68); 94.1% match greedy (64/68); 0% match neither. On decisions where planning matters, Opus behaves almost identically to a greedy agent.

CoT verification: Manual inspection of all 68 hard decisions found 2 instances of genuine planning (2.9%) and 2 borderline cases. The remaining 64 show no forward-looking reasoning.

J-value comparison:

Agent	Mean J	vs. Random (p)	vs. Greedy (p)
Optimal	$+1.30$	—	—
Greedy	$+1.19$	—	—
Opus	-0.16	< 0.0001	$= 0.0001$
Random	-1.81	—	—

Opus significantly outperforms random but significantly underperforms greedy.

6.3 Failure Modes

Two qualitative patterns emerge (Figures E, F). **Pathological alternation** (e.g., seed 8): Opus switches at nearly every step, incurring excessive costs—reacting to noise rather than signal. **Excessive caution**: Opus never switches on, missing positive-signal opportunities. Both are consistent with the absence of forward-looking reasoning.

6.4 Temporal Distribution and Information Availability

Hard decisions are spread across the full trajectory (15% early, 53% mid-episode, 32% late), with the average hard decision at $t = 11$. At that point, the model has observed sufficient signal history to estimate persistence—yet it follows greedy logic on 94.1% of hard decisions. This rules out an information-deficit explanation: the failure is in dynamic reasoning, not data availability.

---

7. Discussion

7.1 A Single Gap: Planning

The two-bucket analysis isolates a single deficit: on the 68 decisions requiring forward-looking reasoning, Opus follows greedy logic almost perfectly. Comprehension is near-ceiling (99.3%); the only gap is dynamic planning.

A manual scaffolded test (pre-computing option values, flagging signal trends, prompting persistence reasoning) yielded 25/25 correct decisions including the hard one. However, this scaffold essentially performs the planning for the model—the model followed instructions, not planned independently. The honest interpretation: planning capability is latent but not spontaneously available, confirming the gap RLVR training should target.

7.2 Implications for RLVR Training

The near-zero planning rate provides a concrete, measurable target for RLVR training. The gym's exact ground truth and regret-normalized scoring are directly compatible with GRPO. The hard-bucket planning rate is the primary metric to monitor during training; easy accuracy is already at ceiling.

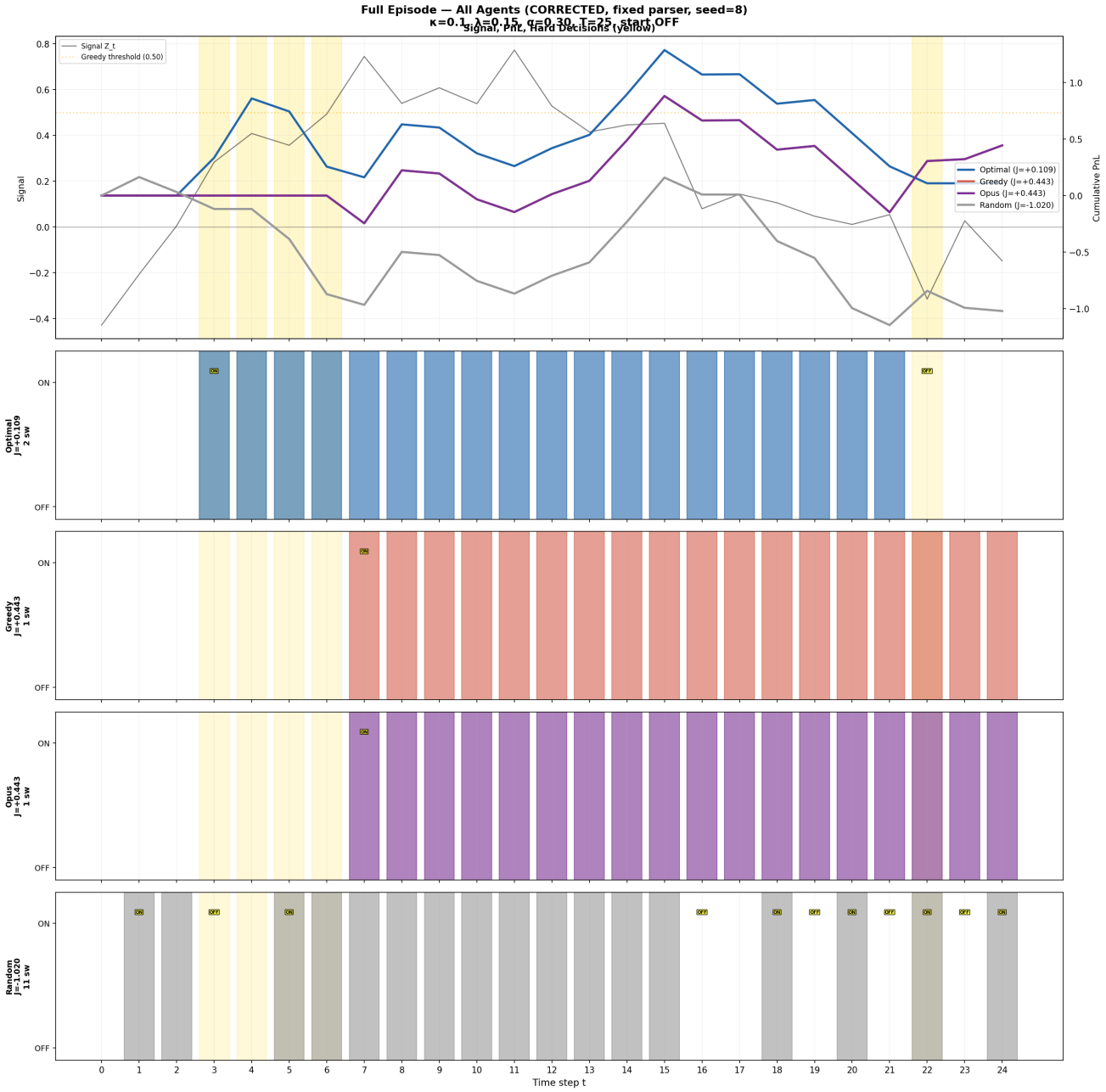


Figure 4: Figure E — Full Episode: Opus Alternating (seed=8)

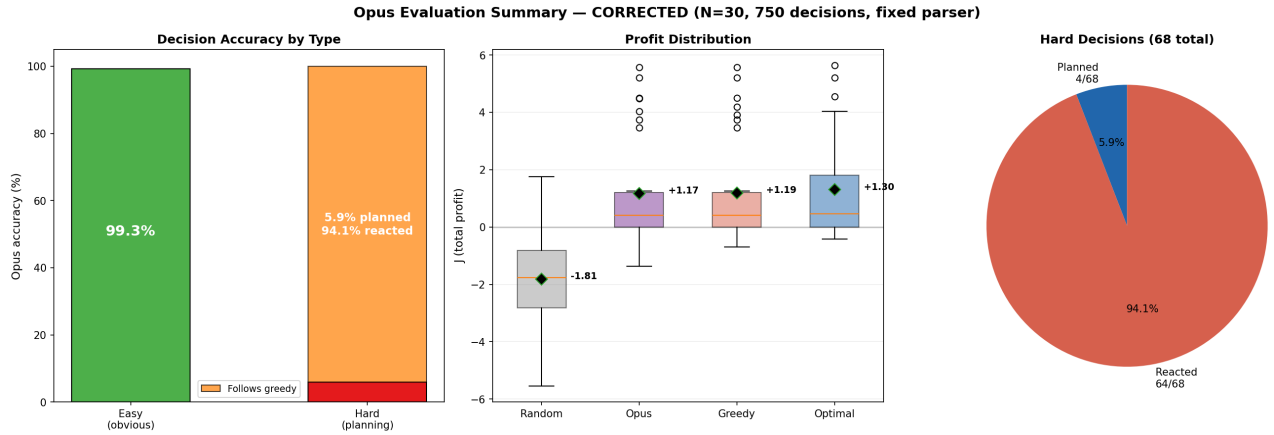


Figure 5: Figure G — Opus Evaluation Summary

8. Limitations

The gym contains a single problem type (binary switching on scalar OU signal). The planning advantage ceiling is modest (~22% of states, ~10% of trajectory decisions). We evaluate only one model (Claude Opus 4). The OU process is stationary, unlike real markets. We provide no RLVR training results—only infrastructure and baseline. The LLM does not receive κ, creating an information asymmetry with the solver; a POMDP formulation would make the comparison strictly fair. LLM performance is sensitive to prompt phrasing.

---

9. Conclusion

We have presented the first verifiable RLVR environment for financial planning, with exact optimal solutions via Bellman backward induction. Signal persistence (κ) dominates the parameter landscape; the disagreement zone is analytically transparent. Evaluating Claude Opus 4, we find near-perfect comprehension but near-zero planning: the model achieves 99.3% on easy decisions and 5.9% on hard decisions where forward-looking reasoning is required. Forward-looking financial planning is not an emergent capability of current frontier LLMs. The gym is open-source with TRL integration and designed for extension within the strategy-relative framework.

References

Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *J. Risk*, 3(2):5–39.

Bilokon, P. (2026). Strategy-relative market regimes as filtration compressions. SSRN 6504227.

Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019). Deep hedging. *Quant. Finance*, 19(8):1271–1291.

DeepSeek-AI et al. (2025). DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv:2501.12948.

Liu, X.-Y. et al. (2022). FinRL-Meta: Market environments and benchmarks for data-driven financial RL. *NeurIPS 2022 Datasets and Benchmarks*.

Merton, R. C. (1969). Lifetime portfolio selection under uncertainty. *Rev. Econ. Stat.*, 51(3):247–257.

Shao, Z. et al. (2024). DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. arXiv:2402.03300.

Stojanovski, D. et al. (2025). Reasoning Gym: Reasoning environments for RL with verifiable rewards. *NeurIPS 2025 Spotlight*. arXiv:2505.24760.

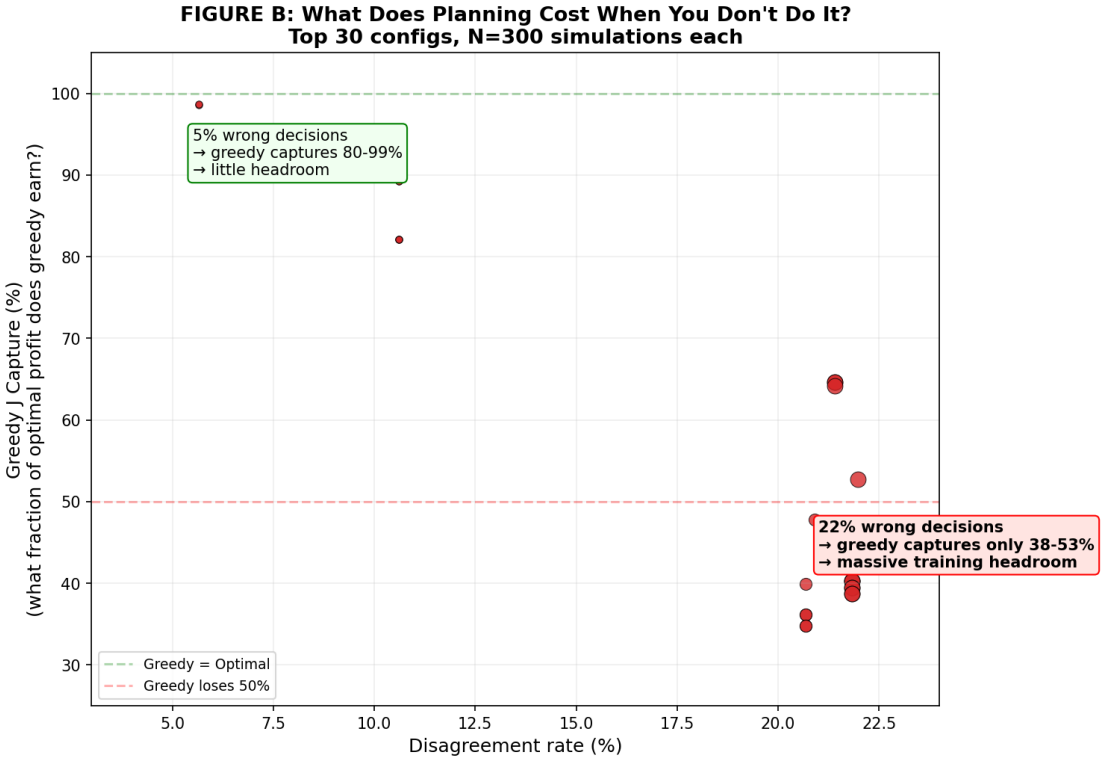


Figure 6: Figure B — Cost of Not Planning

Appendix

A. Aggregate Evaluation Statistics

Metric	Value
Total instances	30
Total decisions	750
Easy / Hard	682 / 68
Easy accuracy	99.3% (677/682)
Hard: matches optimal	5.9% (4/68)
Hard: matches greedy	94.1% (64/68)
CoT-verified planning	2.9% (2/68)
Mean J: Opus / Optimal / Greedy / Random	-0.16 / $+1.30$ / $+1.19$ / -1.81

B. Additional Figures

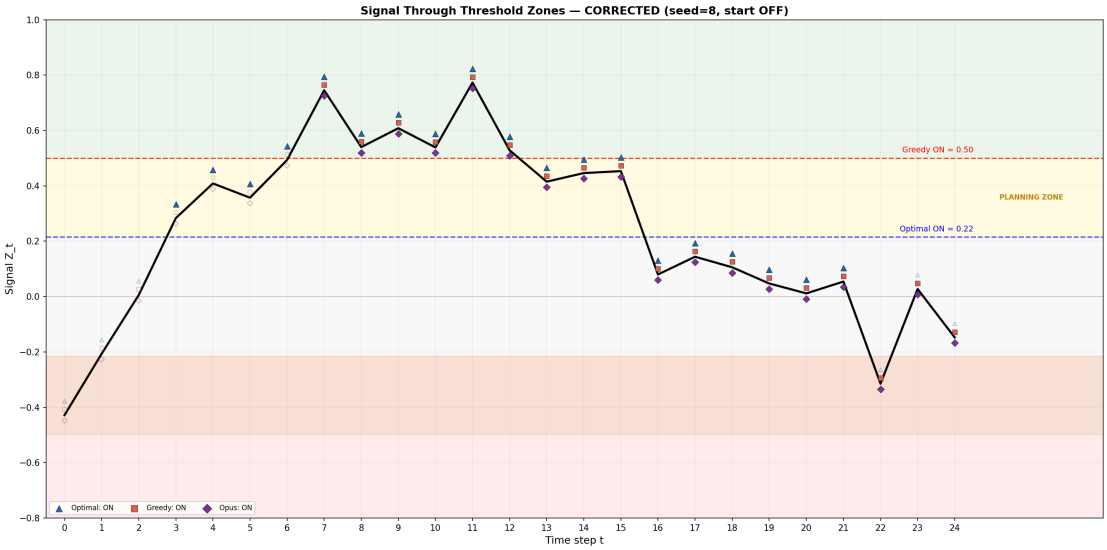


Figure 7: Figure H — Signal Path Through Threshold Zones

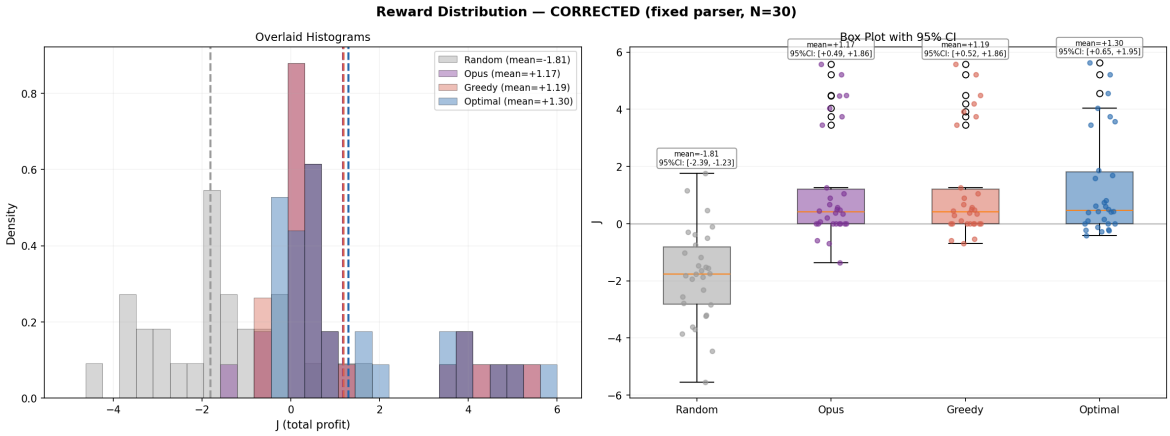
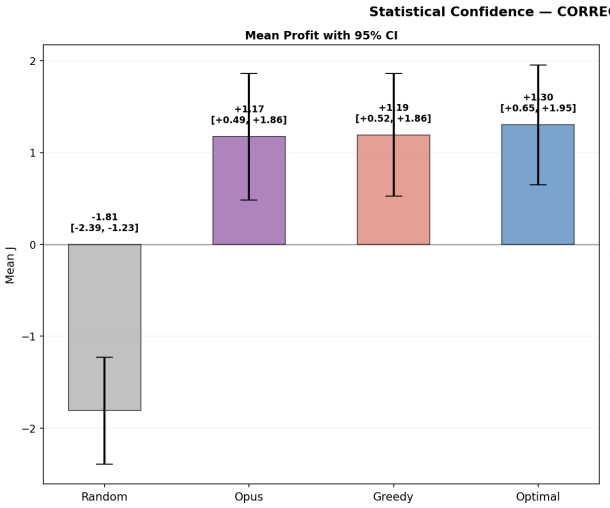


Figure 8: Figure I — Reward Distribution



Statistical Tests (corrected)

Comparison	Mean diff	t-stat	p-value	Significant?
Opus vs Random	+2.984	6.49	0.0000	YES
Opus vs Greedy	-0.018	-0.04	0.9708	no
Opus vs Optimal	-0.127	-0.26	0.7936	no
Greedy vs Optimal	-0.109	-0.23	0.8195	no
CORRECTED RESULTS				
Easy accuracy	99.3%	677/682		
Hard: plans	5.9%	4/68		
Hard: greedy	94.1%	64/68		
CoT-verified	2.9%	2/68		

Figure 9: Figure K — Statistical Confidence (CIs and t-tests)